

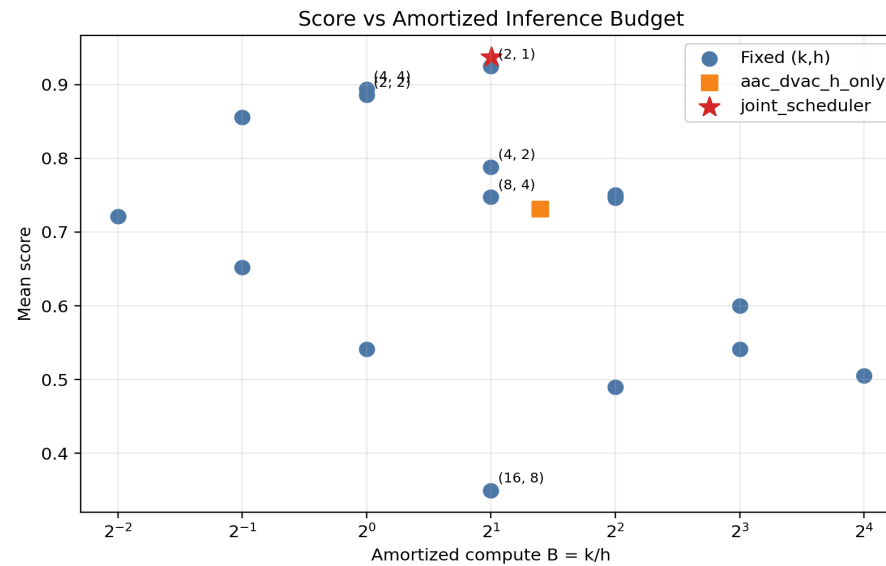
Joint Compute-Control Budgeting

Iso-compute denoising and replanning for Diffusion Policy-style control

- Final project: reproduce a robotics paper and add an improvement idea
- Base paper: Diffusion Policy, RSS 2023
- Task: Push-T-style closed-loop contact benchmark

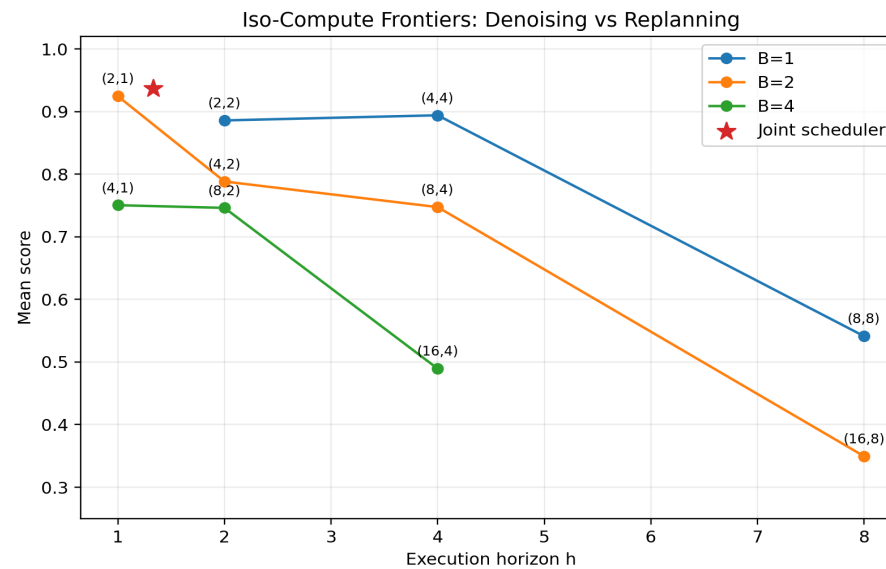
Motivation

- Diffusion Policy is accurate but iterative denoising is expensive
- Replanning improves reactivity but increases policy calls
- Deployment needs a fixed per-step inference budget



Key Question

- Should compute be spent on larger k or smaller h ?
- k : denoising steps per policy call
- h : executed controls before replanning
- $B \approx k / h$ is the iso-compute budget



Reproduction Setup

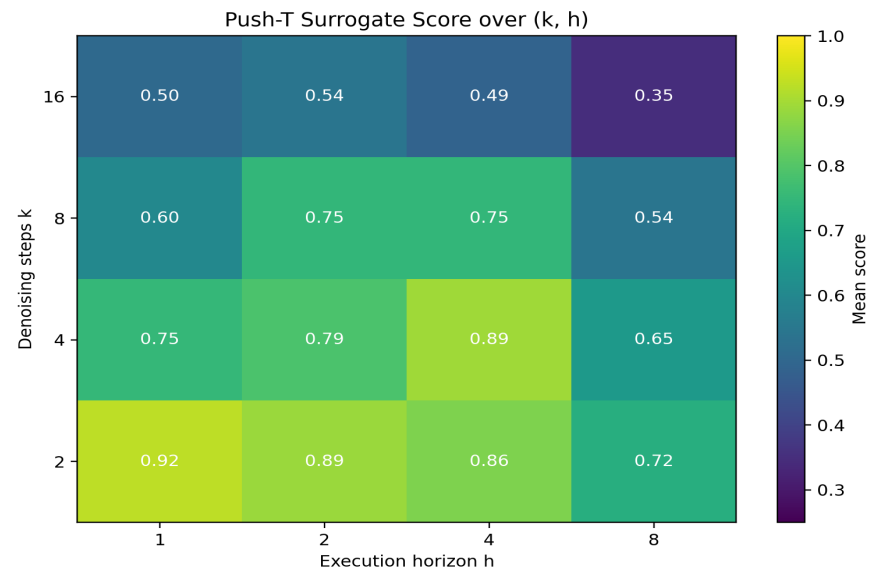
- Official source acquired via GitHub codeload archive
- Official Push-T checkpoint downloaded: 1044185793 bytes
- 50-seed official runner score: 0.919
- Compatibility patches needed for Python 3.12 / Gym 0.26

LeRobot PushT Checkpoint

- Model card 500-episode max reward: 0.955
- Model card success rate: 65.4%
- Local safetensors loads and runs in real gym-pusht
- Local smoke: max reward 0.627, 100 denoising steps
- Caveat: mirror file did not pass HF LFS SHA check

Static Grid

- Grid: k in $\{2,4,8,16\}$, h in $\{1,2,4,8\}$
- Contact-rich pushing makes equal-compute points behave differently
- The best fixed $B=2$ score point is $(2,1)$



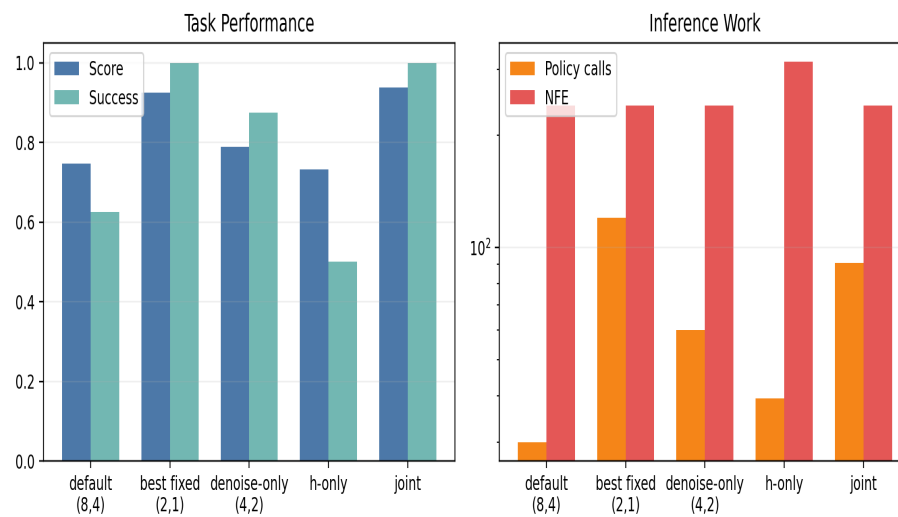
Joint Scheduler

- Training-free rule on the $B=2$ frontier
- Use (4,2) for free-space approach and final alignment
- Use (2,1) during close-contact pushing or unstable progress
- Goal: keep compute fixed while moving reactivity to high-risk phases

Main Results

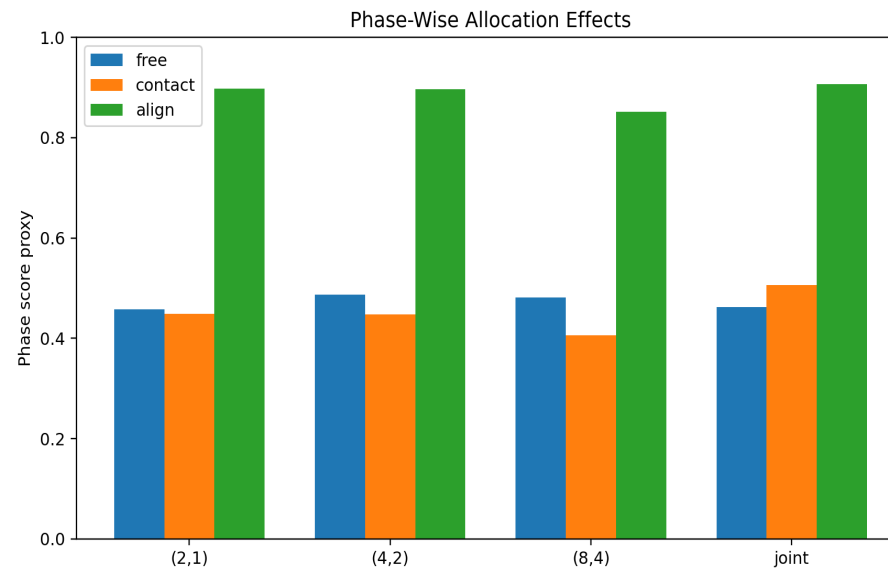
- Joint score: 0.937
- Best fixed B=2 score: 0.925
- Policy calls: 90.8 vs 120.0
- Smoothness cost: 0.271 vs 0.444

Main Baseline Comparison under $B \approx 2$



Phase-Wise Analysis

- Free-space approach tolerates longer chunks
- Contact pushing needs frequent replanning
- Near-goal alignment benefits from smoother denoised chunks



Limitations

- Improvement numbers are from the surrogate, not official benchmark rollouts
- Official run uses compatibility patches and scores below checkpoint filename
- LeRobot checkpoint path is executable but not SHA-verified
- $B=k/h$ is an approximate compute model
- No real-robot validation

Takeaways

- Denoising and replanning are coupled deployment decisions
- Best allocation changes by task phase
- A simple joint scheduler can improve the score-smoothness tradeoff under the same budget